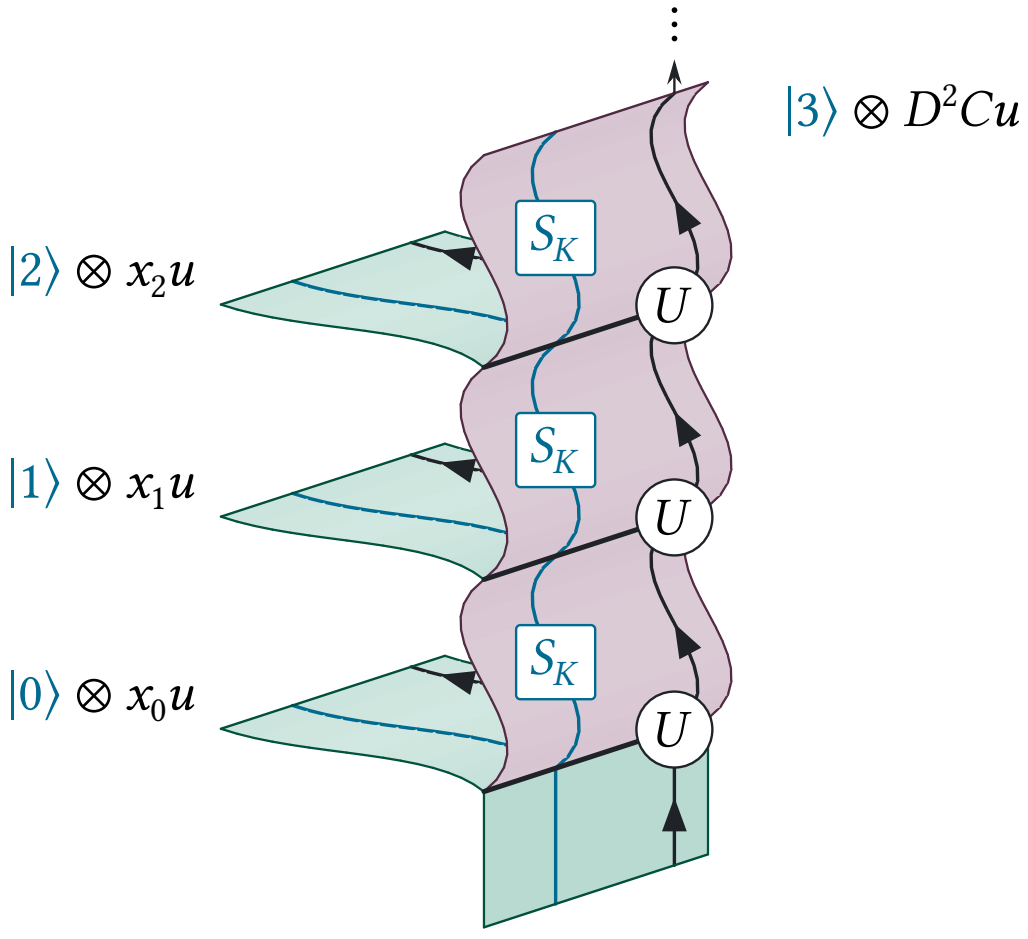
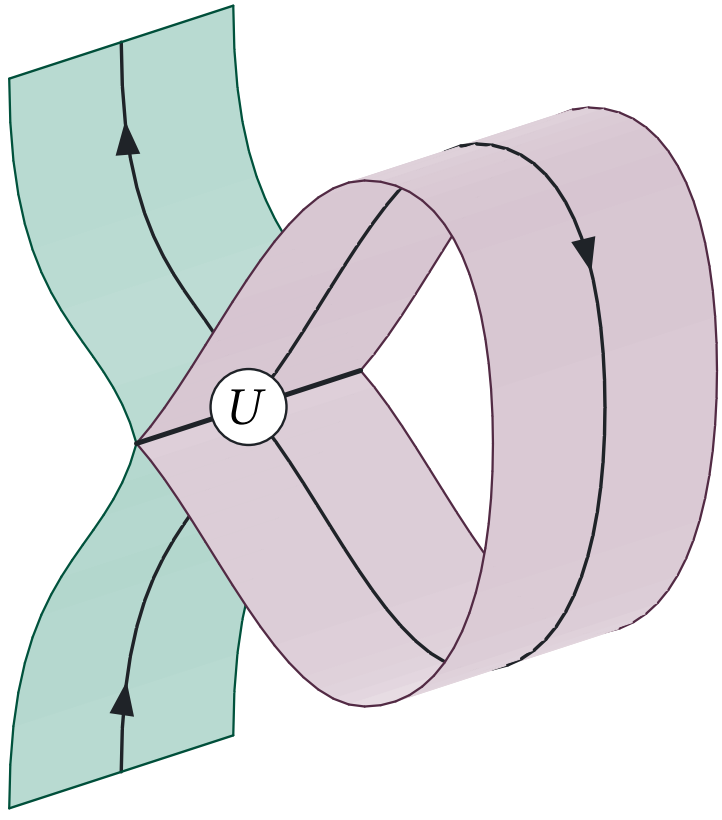
(a) Keep the count

(b) Coherent feedback



$\mathrm{alloc}_0(u) = |0\rangle \otimes u$

$Lu = (x_0 + x_1 + x_2 + \cdots)u$



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